Sabastian Asuo-Darko

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Professional Summary

Research-focused Biomedical Engineer with First-Class Honours and a peer-reviewed publication in pressure ulcer prevention, underpinning future smart medical device development. Applied computational modelling (ANSYS, MATLAB, SolidWorks) to study pulmonary mechanics in ventilator-induced lung injury for undergraduate thesis, translating physiological insight into simulation-driven design. Combines clinical engineering experience with computational biomechanics expertise to advance medical device innovation from research to real-world impact.

Curious and coachable professional

Research Interests

- Computational Biomechanics & Physiological Modelling
- Medical Device Design, Simulation & Validation
- Respiratory Mechanics & Ventilator-Induced Lung Injury (VILI)
- Biomedical Systems Analysis & Translational Engineering

Publications & Projects

Sept 2023 – Jan 2024

Causal Factors and Their Effects on the Human Body for the Design of a Smart Bedsore Prevention System

Co-authored Publication — IJITIS *KNUST, Ghana*

- Co-authored a multidisciplinary study analysing pressure, temperature, and humidity as key contributors to pressure ulcer formation.
- Supported system design, data analysis, and modelling to inform the development of a smart bedsore prevention device.
- Research addresses a clinical challenge affecting 2.5M+ patients annually, with projected reduction potential of up to 40% in high-risk groups.
- View Publication

Jan 2024 – Aug 2024

Simulating Pulmonary Mechanics

Undergraduate Thesis — Computational Modelling of VILI *KNUST, Ghana*

- Led a two-person research project modelling spatial pressure propagation in lung tissue to study ventilator-induced lung injury (VILI).
- Designed anatomically inspired 3D lung models and ran ANSYS simulations to evaluate tissue response under varying ventilation pressures.
- Analysed simulation outputs to identify stress distribution trends relevant to ventilator control optimisation.
- Findings contribute to improved mechanical ventilation strategies for ICU patients requiring respiratory support.
- View Documentation

Jan 2022

Design and Finite Element Analysis of an Above-Knee Prosthetic Leg

Biomedical Engineering Design Project *KNUST, Ghana*

- Designed a low-cost above-knee prosthetic leg using the biomedical engineering design process.
- Conducted market analysis, functional decomposition, and QFD to define engineering specifications.
- Modelled prosthetic components and full assembly in SolidWorks.
- Performed FEA to assess stress, displacement, and strain under physiological loading.
- Evaluated material selection and proposed validation tests for safety and durability.
- View Documentation

June 2024

Computational Design and Testing of Sports Equipment

Engineering Simulation Project *KNUST, Ghana*

- Applied computational modelling techniques to analyse stress distribution and performance characteristics.
- Used simulation-driven insights to guide iterative design improvements.
- View Documentation

Industry and Clinical Experience

May 2026 – Present

Skyler Healthcare

Field Service Engineer (Freelance) *Accra, Ghana*

- Installed and commissioned haematology analysers at various healthcare facilities.
- Performed preventive and corrective maintenance, maintaining 100% device uptime.
- Delivered hands-on clinical training for laboratory and clinical staff.
- Lead technical presentations and product comparisons during tenders and procurement.
- Manage customer accounts and support contract renewals and equipment upgrades.

Jan 2026 – May 2026

Machine Medics
Field Service & Sales Engineer (Contract) *Oyarifa, Ghana*

- Installed and commissioned 15+ immunoassay analysers, ECG machines, and chemistry analysers.
- Performed preventive and corrective maintenance on haematology and chemistry systems.
- Delivered hands-on clinical training for laboratory and clinical staff.
- Managed end-to-end technical service operations and key accounts.
- Resolved open field cases and maintained operational continuity for critical clients.
- Lead technical presentations, proposals, and product comparisons during tenders.

Oct 2024 – Jan 2026

Sysmex West and Central Africa
Field Service Engineer (Asst.) *Cantonments, Ghana*

- Conducted site readiness assessments for 20+ haematology and urinalysis analysers, reducing setup delays by 30%.
- Installed and commissioned diagnostic systems to full operational readiness within one day per site.
- Carried out monthly preventive maintenance on 20+ analysers, reducing downtime by 20%.
- Sustained 95%+ device uptime across assigned sites; documented all service activity.
- Trained clinical and laboratory staff on equipment operation and troubleshooting.

Sept 2023 – Dec 2023 & Sept 2021 – Dec 2021

UGMC & Ministry of Health — Biomedical Engineering Unit
Biomedical Engineering Intern *Legon & Ridge, Ghana*

- Performed routine maintenance on 20+ medical devices monthly, reducing failure rates by 25%.
- Repaired 10+ critical devices (suction units, infusion pumps), preventing OR and ward delays.
- Supported safe usage of 5+ surgical systems, including laser and electrosurgical units.
- Tested and validated 30+ medical devices across 10 hospitals, ensuring 100% compliance.
- Logged device performance data, improving asset tracking efficiency by 25%; supported national asset verification across 5+ regions.

More info on industry experience here: Professional Portfolio

Education

Jan 2021 – Nov 2024

Kwame Nkrumah University of Science & Technology (KNUST)
BSc. Biomedical Engineering — First Class Honours *Kumasi, Ghana*

CWA 77.75/100 · Top 10% of cohort · Winner, BMESS Day of Code (July 2023) · Deputy Judicial and Electoral Committee Head, BMESS (2022–2024)

Volunteer Experience

May 2026 – Present

Gramic Foundation
Video Editor (Volunteer) *Remote*

- Create and edit video content for community outreaches, advocacy campaigns, and mentorship programs.
- Produce educational and promotional materials to support literacy initiatives and environmental sustainability awareness.
- Collaborate with the communications team to develop engaging visual content for social media and digital platforms.
- Contribute to the foundation's mission of bridging the literacy gap and inspiring lifelong learning in less endowed communities.

Software Competencies

Engineering & Simulation
ANSYS, MATLAB, LabVIEW

Programming & Data
Python, C, C++, SQL, AI & ML

Design & Analysis
SolidWorks, AutoCAD, Blender

Productivity & Reporting
Microsoft Office Suite, Google Workspace, ZohoOne

Other Tools
Git, ImageJ, Power BI, Adobe Suite

Certifications

Medical Device Essentials — SternMed GmbH · WHO Covid-19 Respiratory Equipment · Sysmex Caresphere User Operation Training · ALX AI Career Essential Course · Professional Development: IT Security & AI

Languages: English (Native) · French (Conversational)